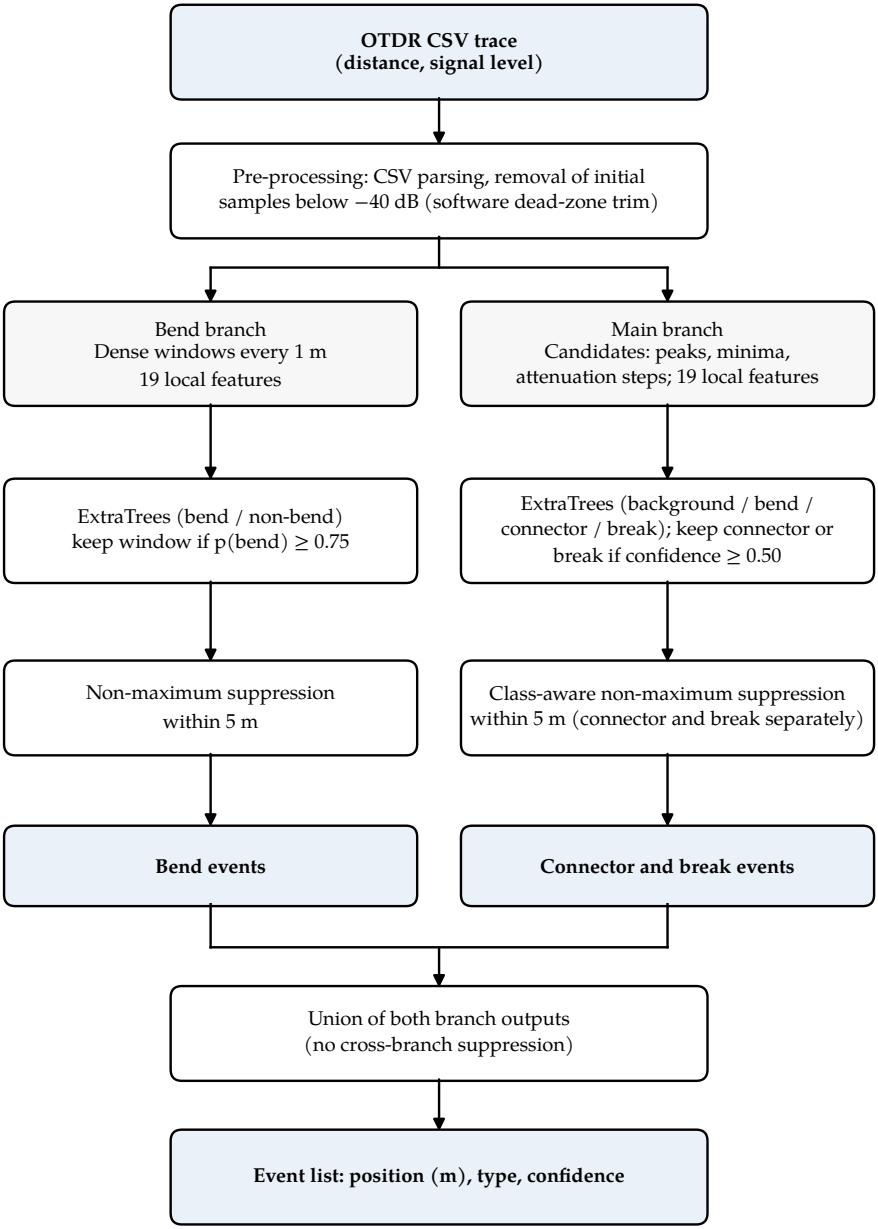
OTDR CSV trace
(distance, signal level)
Pre-processing: CSV parsing, removal of initial samples below −40 dB (software dead-zone trim)
Bend branch
Dense windows every 1 m
19 local features
Main branch
Candidates: peaks, minima, attenuation steps; 19 local features
ExtraTrees (bend / non-bend)
keep window if p(bend) ≥ 0.75
ExtraTrees (background / bend / connector / break); keep connector or break if confidence ≥ 0.50
Non-maximum suppression within 5 m
Class-aware non-maximum suppression within 5 m (connector and break separately)
Bend events
Connector and break events
Union of both branch outputs
(no cross-branch suppression)
Event list: position (m), type, confidence